

Lattice Resonances Induced by Periodic Vacancies in Arrays of Nanoparticles

Lauren Zundel, Asher May, and Alejandro Manjavacas*

Cite This: *ACS Photonics* 2021, 8, 360–368

Read Online

ACCESS |



Metrics & More



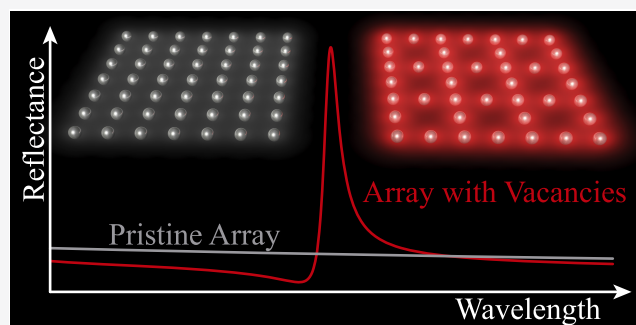
Article Recommendations



Supporting Information

ABSTRACT: Periodic arrays of nanoparticles are capable of supporting lattice resonances, collective modes arising from the coherent interaction of the particles in the array. These resonances, whose spectral position is determined by the array periodicity, are spectrally narrow and lead to strong optical responses, making them useful for a wide range of applications, from nanoscale light sources to ultrasensitive biosensors. Here, we report that, by removing particles from an array in a periodic fashion, it is possible to induce lattice resonances at wavelengths commensurate with the periodicity of these vacancies, which would otherwise not be present in the system. Using a coupled dipole approach, we perform a comprehensive analysis of how the properties of these vacancy-induced lattice resonances depend on the array periodicity, the particle size, and the number of vacancies per unit of area. Furthermore, we find that these lattice resonances have a subradiant character and originate from the symmetry breaking introduced in the unit cell by the presence of the vacancies. Finally, we investigate a potential implementation of an array with vacancies made of nanocylinders embedded in a homogeneous dielectric environment. The results of this work serve to advance our understanding of lattice resonances and provide an alternative method for controlling the optical response of periodic arrays of nanostructures.

KEYWORDS: lattice resonances, vacancy, periodic arrays, nanoparticle arrays, plasmons, plasmonic crystals



Metallic nanostructures are known to support surface plasmons, collective oscillations of their conduction electrons, characterized by strong optical responses and subwavelength field confinement.^{1–3} Thanks to these properties, there has been a tremendous effort dedicated to designing and investigating the optical response of plasmonic nanostructures with different geometrical properties and arrangements. A particularly interesting example is the placement of metallic nanostructures in a two-dimensional periodic array, which can give rise to collective responses known as lattice resonances.^{4–10} These excitations, which occur at wavelengths commensurate with the periodicity of the array, are the result of the coherent interaction between the localized plasmons of the array constituents.^{11,12} Due to their collective nature, lattice resonances produce strong optical responses, both in the far-^{13–15} and near-field,^{16–19} that are spectrally narrow and result in record-high quality factors for metallic systems.^{20–25} Such extraordinary properties make periodic arrays of nanostructures excellent candidates for a variety of applications such as optical sensing,^{26–29} nanoscale light emission,^{30–38} lensing,³⁹ second-harmonic generation,⁴⁰ and color generation,^{41–43} as well as platforms to explore new physical phenomena.^{44–46}

Recently, there has been a growing interest in investigating the optical response of periodic arrays of metallic nanostructures built from the repetition of unit cells containing more

than one particle,^{15,47–57} that is, associated with a non-Bravais lattice. As expected, these systems, being more complex than arrays composed of single-particle unit cells, display much richer optical responses.^{25,52,58–60} One possible way to create an array with a multiparticle unit cell is to start from a single-particle array and remove particles in a periodic fashion. This results in an array that can be seen as the subtraction of two square arrays with different periods. Since the resulting system is still periodic, it is expected to support lattice resonances. However, the following questions arise: What are the properties of these lattice resonances and how is their origin connected to the presence of the periodic vacancies? In order to answer these questions, we employ a coupled dipole approach to perform a comprehensive analysis of how the particle size, array periodicity, and density of vacancies determine the optical response of the array. In particular, we find that the lattice resonances of these systems appear farther from the corresponding Rayleigh anomalies and result in

Received: October 30, 2020

Published: January 6, 2021



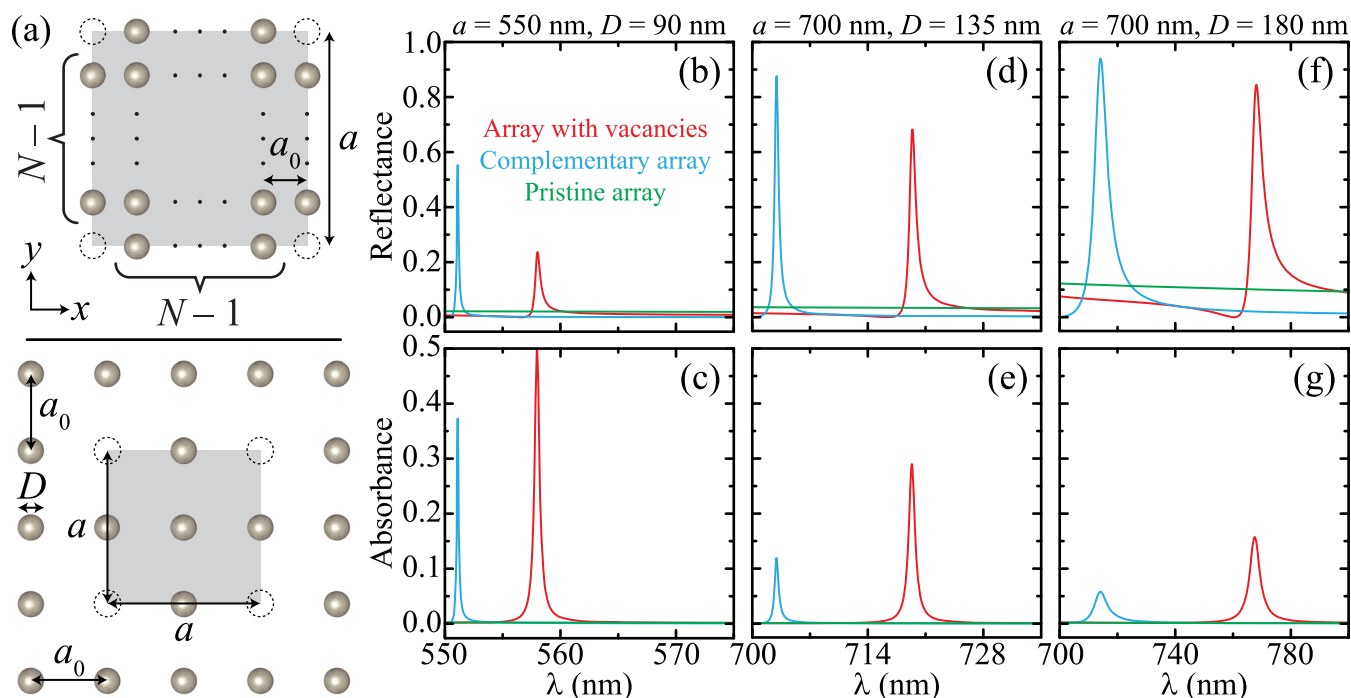


Figure 1. Lattice resonances induced by the introduction of periodic vacancies in a periodic array of nanoparticles. (a) Schematics of the system under consideration, which is built by removing, from a square array of period a_0 , the particles at the corners of each block of $(N + 1) \times (N + 1)$ particles. This results in a new square array of period $a = Na_0$, with a unit cell containing $N^2 - 1$ particles (gray box), as shown in the upper panel. The specific case for $N = 2$ is shown in the lower panel. (b, c) Reflectance (b) and absorbance (c) spectra around the first lattice resonance of an array with $N = 2$, $a = 550$ nm, and $D = 90$ nm. The red curve corresponds to the array with vacancies, as the one depicted in (a), while the blue one represents the complementary array, consisting of the array formed by the removed particles. The green curve shows the response of the pristine square array without vacancies. (d–g) Same as (b, c), but for $a = 700$ nm and either $D = 135$ nm (d, e) or $D = 180$ nm (f, g).

smaller reflectance and larger absorbance than those of a single-particle array with the same periodicity. Furthermore, to fully understand the nature of these lattice resonances, we investigate the dipoles induced in each of the particles. By doing so, we find that these modes have a subradiant character and their origin can be traced back to the symmetry breaking introduced in the unit cell by the presence of the vacancies. Finally, we analyze a potential implementation of these results using arrays of nanocylinders embedded in a homogeneous dielectric environment.

Periodic vacancies have been previously studied in the context of photonic crystals.^{61,62} The results of this work provide an understanding of their effect on the response of periodic arrays of metallic nanoparticles and therefore serve to increase the potential of these systems for technological applications such as ultracompact biosensors and nanoscale light sources.

RESULTS AND DISCUSSION

The system under consideration, which is depicted in the upper schematic of Figure 1a, is built as follows: we start with an array of spherical silver nanoparticles of diameter D , arranged in a square lattice of period a_0 , from which, then, we remove the nanoparticles at the four corners of each block of $(N + 1) \times (N + 1)$ particles. This results in a new square array of period $a = Na_0$ with a unit cell containing $N^2 - 1$ particles and one vacancy. The geometry of the simplest case, for which $N = 2$, is shown in the lower schematics of panel (a). Assuming that the nanoparticle diameter D is much smaller than a_0 and the relevant wavelengths, we can describe the optical response of this array using the coupled dipole model (CDM).^{4,6,10,14,52}

Under this approximation, each spherical nanoparticle is modeled as a point electric dipole $\mathbf{p}$ with polarizability α , which we compute from the dipolar Mie scattering coefficient⁶³ using tabulated dielectric function data from ref 64 (see Figure S1 of the Supporting Information for selected examples of the polarizability). Following the approach described in ref 52, the dipole induced in each particle of the array, when illuminated by an external field of amplitude $\mathbf{E}_{i,\mu}$ is

$$\mathbf{p}_{i,\mu} = \alpha \mathbf{E}_{i,\mu} + \alpha \sum_{\nu}' \sum_{j=1}^{N^2-1} \mathbf{G}_{ij,\mu\nu} \mathbf{p}_{j,\nu} \quad (1)$$

Here, we use latin and greek indices to denote, respectively, each particle within the unit cell and the unit cell to which it belongs. Furthermore, the prime symbol on the first summation indicates that the term $\nu = \mu$ is excluded if $i = j$, since a dipole does not interact with itself. For different dipoles, however, their interaction is encoded in the dipole–dipole interaction tensor $\mathbf{G}_{ij,\mu\nu} = [k^2 + \nabla \nabla] e^{ik|\mathbf{T}_\mu + \mathbf{r}_i - \mathbf{T}_\nu - \mathbf{r}_j|} / |\mathbf{T}_\mu + \mathbf{r}_i - \mathbf{T}_\nu - \mathbf{r}_j|$, where $\mathbf{r}_i$ indicates the position of the particle i within its unit cell, $\mathbf{T}_\mu - \mathbf{T}_\nu$ is the distance between unit cells μ and ν , and $k = 2\pi/\lambda$ is the wavevector, with λ being the wavelength of the light.

Since the arrays considered in this work are perfectly periodic, the solutions of eq 1 satisfy Bloch's theorem and can be written as $\mathbf{p}_{i,\mu} = \mathbf{p}_i(\mathbf{k}_\parallel) e^{ik_\parallel \mathbf{T}_\mu}$. In this expression, $\mathbf{k}_\parallel$ is the component of the wavevector parallel to the array, which, throughout this work, is always zero since we assume the arrays to be excited at normal incidence. Furthermore, without the loss of generality, we choose the incident electric field to be

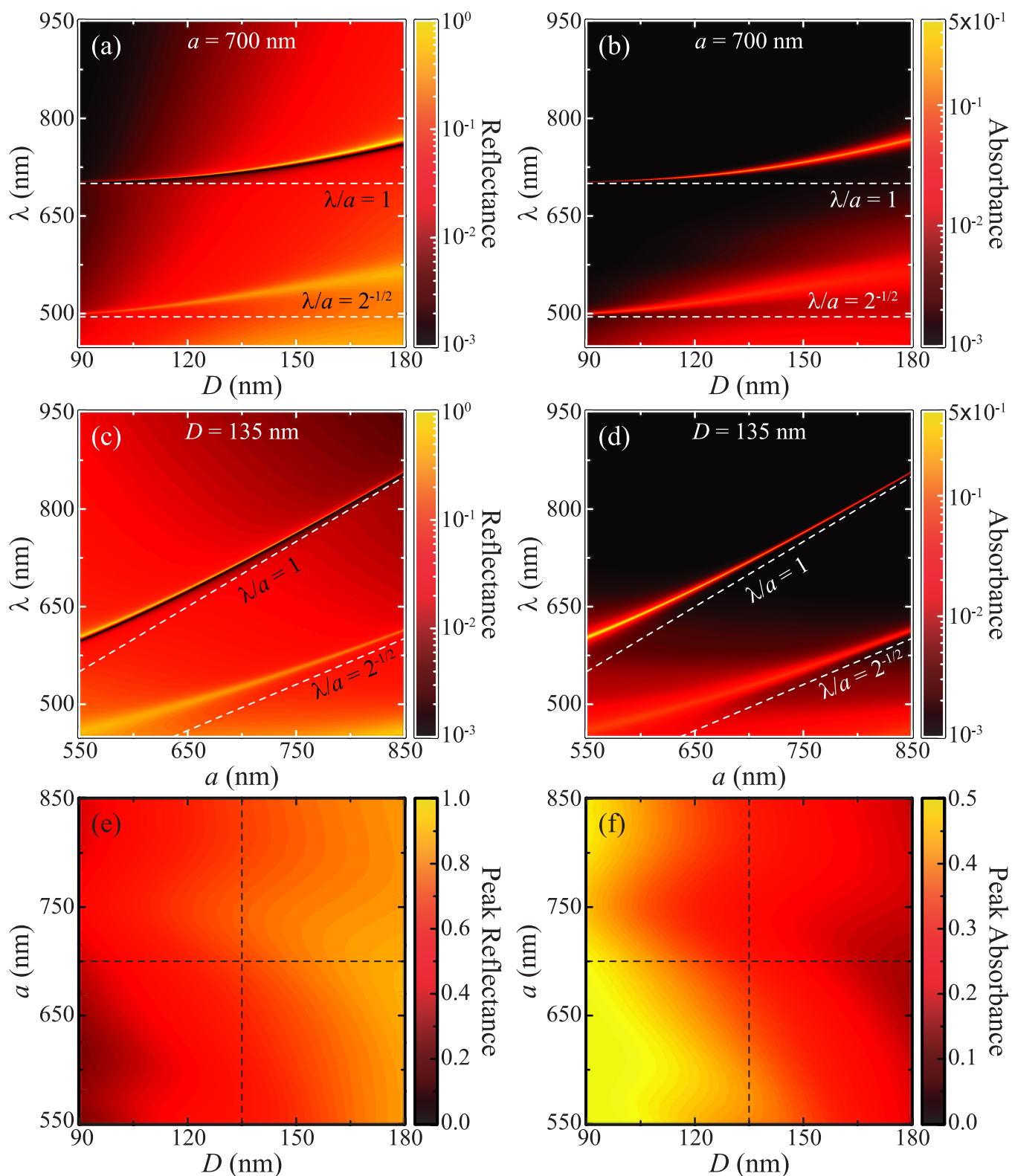


Figure 2. Role of the array period and nanoparticle diameter on the lattice resonances of arrays with vacancies. (a, b) Spectral dependence of the reflectance (a) and absorbance (b) for arrays with different nanoparticle diameter D and constant period $a = 700$ nm. The white dashed lines indicate the Rayleigh anomalies at $\lambda = a$ and $\lambda = a/\sqrt{2}$. (c, d) Same as (a, b), but for a fixed diameter $D = 135$ nm and varying period a . (e, f) Maximum value of the reflectance (e) and absorbance (f) as a function of D and a . The black dashed lines indicate the slices corresponding to the cases analyzed in (a)–(d).

polarized along the x -axis, that is, $\mathbf{E}_{i,\mu} = \mathbf{E}_i = E_0 \hat{\mathbf{x}}$. Then, the

$$\mathbf{M} = \alpha^{-1} \mathbf{I} - \mathcal{G} \quad (2)$$

solution of eq 1 is given by $\mathbf{p}_i = \sum_{j=1}^{N^2-1} \mathbf{M}_{ij}^{-1} \mathbf{E}_j$, with

Here, I is the identity matrix of dimension $3(N^2 - 1) \times 3(N^2 - 1)$, and $\mathcal{G}$, whose components are defined as $\mathcal{G}_{ij} = \sum_{\nu} \mathbf{G}_{ij,\nu,0}$, is known as the lattice sum.^{6,52} This quantity determines the collective response of the array and, in particular, the properties of the lattice resonances, which appear at the poles of $\mathbf{M}^{-1}$. These poles are located near the Rayleigh anomalies of the system at which the real part of $\mathcal{G}$ diverges. At normal incidence, the wavelengths of the Rayleigh anomalies are $\lambda_{\mathbf{q}} = 2\pi/|\mathbf{q}|$, where $\mathbf{q}$ are the reciprocal lattice vectors of the array $\mathbf{q} = 2\pi(m\hat{x} + n\hat{y})/a$, with m and n being integers.^{10,14}

Once we know the induced dipoles, the reflectance and absorbance of the array can be calculated from the power that it transmits and reflects, normalized to the incident power. We do this in Figure 1b,c for an array with vacancies with $N = 2$, $a = 550$ nm, and $D = 90$ nm (red curves). Examining these results, which are in excellent agreement with rigorous solutions of Maxwell's equations obtained using a finite element method (FEM) solver (see Figure S2 of the Supporting Information), we clearly observe a peak at a wavelength of 558 nm. This resonance, which we identify as the first lattice resonance of the array (i.e., the one associated with the first Rayleigh anomaly), results in a moderate reflectance ~ 0.24 and an absorbance that saturates the maximum value achievable for a perfect two-dimensional system, 0.5.⁶⁵ It is interesting to compare this response with that of the complementary array, which is shown by the blue curve. We define the complementary array as the array formed by the removed particles, which therefore has the same period a , but only one particle per unit cell. Similar to the array with vacancies, the complementary array also displays a lattice resonance, although it occurs at a wavelength of 551 nm, which is closer to the first Rayleigh anomaly located at $\lambda = a = 550$ nm. Interestingly, the lattice resonance of this system results in a larger reflectance but smaller absorbance than that of the array with vacancies. To complete the comparison, we also plot, using green curves, the response of the pristine array (i.e., the original array without vacancies). As expected from the fact that its period is $a_0 = 275$ nm, that is, half of that of the other two arrays, this system does not support a lattice resonance in the spectral range under consideration. For completeness, Figure S3 of the Supporting Information displays the optical response of the arrays across a wider spectral range.

The array with vacancies considered in Figure 1b,c has an optical response that is dominated by the absorbance. However, it is well-known that, by increasing the ratio of the nanoparticle size to the array periodicity, it is possible to obtain a system that, instead, has a larger reflectance.^{14,19,25} Interestingly, this behavior is true not only for the lattice resonances of single-particle arrays, but also for those of the arrays with vacancies. This can be seen in Figure 1d,e, where we repeat the calculations from panels (b) and (c), but for $a = 700$ nm and $D = 135$ nm. Clearly, the lattice resonance of the array with vacancies, which occurs at a wavelength 719 nm, results in a significantly larger reflectance (~ 0.68) but a smaller absorbance (~ 0.29). The reflectance further grows to ~ 0.84 when the diameter is increased to $D = 180$ nm, which also results in the reduction of the absorbance to ~ 0.16 , as shown in Figure 1f and g. In addition, this increase in the particle size shifts the lattice resonance farther away from the Rayleigh anomaly as compared with the case of Figure 1d and e. As expected, the response of the corresponding complementary arrays follow the same trends, as seen from the blue curves,

although the lattice resonances of these systems always appear closer to the Rayleigh anomaly, with a higher reflectance and smaller absorbance. On the other hand, the pristine arrays do not show any resonance in this spectral range for either case.

The results shown in Figure 1 confirm that the introduction of vacancies to a periodic array of nanoparticles induces a lattice resonance, whose properties depend on both the size of the particles D and the period of the array a . We can gain further insight into the role played by these two parameters by examining eq 2. In particular, the lattice resonances correspond to poles of $\mathbf{M}^{-1}$, whose spectral position and properties are determined by the interplay between the inverse of the particle polarizability α^{-1} and the lattice sum $\mathcal{G}$. The former is determined by the properties of the nanoparticles, and hence depends on D , while the latter encodes the information of the lattice geometry and, consequently, depends on a . Therefore, for a fixed a , the increase of D results in a smaller value of α^{-1} , which shifts the lattice resonance away from the Rayleigh anomaly to larger wavelengths for which the real part of $\mathcal{G}$ is smaller. This is shown in Figure 2a,b, where we plot, respectively, the reflectance and absorbance of an array with vacancies with $N = 2$ and $a = 700$ nm, as a function of the particle diameter D . Clearly, as D increases, the lattice resonances depart from the corresponding Rayleigh anomaly (dashed white lines) toward larger wavelengths. For the same reasons, if we, instead, keep D fixed and increase a , we expect the lattice resonances to move closer to the corresponding Rayleigh anomalies. This is confirmed in Figure 2c,d, where we plot the reflectance and absorbance for the same array with $N = 2$ and $D = 135$ nm, as a function of a .

To complete our analysis, in Figure 2e,f, we plot the peak values of the reflectance and absorbance for the first lattice resonance as a function of a and D . Examining these results, we can identify a general trend; namely, as D increases, the reflectance grows and the absorbance becomes smaller, which is consistent with the fact that, as the particles become larger, they radiate more efficiently. However, the dependence of the reflectance and absorbance with a is more complicated. In particular, the largest reflectance is achieved for periods in the window 675–750 nm for the range of diameters under consideration. On the other hand, the absorbance reaches its largest values outside of that window: in particular, for periods below 675 nm. This can be attributed to the increased overlap of the lattice resonance with the plasmon resonance of the individual particles, which results in larger material losses in the particles.

So far, we have only considered arrays with $N = 2$, that is, that have a unit cell formed by three particles and one vacancy. However, as was the case with a and D , the number of vacancies per unit of area is expected to have a strong effect on the optical response of the array. To analyze this dependence, in Figure 3, we plot the wavelength of the first lattice resonance λ_{LR} , together with the corresponding peak value of the reflectance (filled dots) and absorbance (empty dots) for different arrays with vacancies. Specifically, in Figure 3a, we analyze arrays with fixed interparticle distance $a_0 = 200$ nm, $D = 100$ nm, and varying N , as indicated by the legend (see Figure S4 of the Supporting Information for the associated spectra). Examining these results, we observe that, as N increases, both the reflectance and the absorbance decrease, although the latter remains close to the theoretical maximum of 0.5 up to $N = 5$, which corresponds to a vacancy for every 24 particles. It is important to notice that the period of these

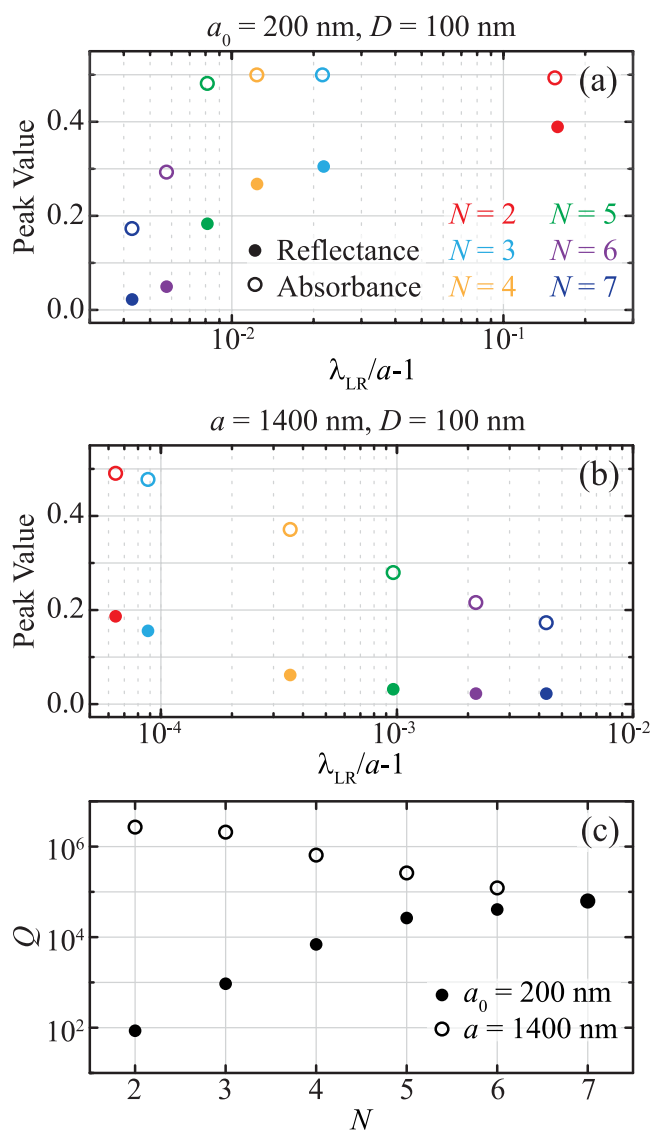


Figure 3. Role of N on the lattice resonances supported by arrays with vacancies. (a) Wavelength of the first lattice resonance λ_{LR} and peak values of the reflectance (filled dots) and absorbance (empty dots) for arrays with vacancies where the interparticle spacing is a constant $a_0 = 200$ nm and N is varied, as indicated by the legend. The array period consequently varies with N as $a = Na_0$. (b) Same as (a), but for arrays with a constant period $a = 1400$ nm and varying interparticle spacing $a_0 = a/N$. In both cases, $D = 100$ nm, and we plot $\lambda_{LR}/a - 1$ instead of λ_{LR} in order to show all of the results on the same scale. (c) Quality factor of the first lattice resonance of all of the arrays from (a) (filled dots) and (b) (empty dots), calculated from the absorbance spectrum.

arrays is $a = a_0N$ and consequently changes with N . Therefore, it is interesting to analyze the behavior of arrays for which, instead, a is fixed and a_0 varies with N as $a_0 = a/N$. The corresponding results, shown in Figure 3b, follow a similar trend as those of Figure 3a, although, in this case, the decrease with N is more gradual (see Figure S5 of the Supporting Information for the associated spectra).

Interestingly, in Figure 3a, the lattice resonance becomes closer to the Rayleigh anomaly as N is increased, whereas, in Figure 3b, the opposite is true. For the arrays of Figure 3a, this behavior can be understood by considering that, as N increases, so does a . Therefore, the wavelength of the lattice

resonance becomes larger and, thus, farther from the plasmon resonance of the individual particles, which results in a smaller particle polarizability. As a result of this, the system requires a stronger collective interaction in order to sustain the lattice resonance or, in other words, a larger value of the lattice sum, which is achieved closer to the Rayleigh anomaly. Similar reasoning explains the behavior of the arrays in Figure 3b. In this case, since a is fixed, the response of the individual particles does not change appreciably as N increases, however, the system becomes more dense (i.e., more particles per unit of area), thus requiring a smaller collective interaction and, therefore, a smaller value of the lattice sum. These behaviors are consistent with previous theoretical predictions for single-particle arrays.¹⁹

Another aspect of the optical response of the arrays with vacancies that is affected by changing N is the linewidth of the lattice resonance (see Figures S4 and S5 of the Supporting Information). This is shown in Figure 3c, where we plot the quality factor $Q = \lambda_{LR}/\Delta\lambda$ of the lattice resonances analyzed in Figure 3a,b. We extract both the wavelength λ_{LR} and the full width at half maximum (fwhm) $\Delta\lambda$ of the lattice resonance from the absorption spectrum and use filled and empty dots to display, respectively, the results for the arrays of Figure 3a,b. Analyzing these results, we observe that, while Q increases with N for the arrays with fixed a_0 (filled dots), it decreases for the ones with fixed a (empty dots). This behavior is tightly connected to the shift of the spectral position of the lattice resonance with N discussed in the previous paragraph. In particular, as we explain above, when N increases (decreases), the lattice resonance of the arrays with fixed a_0 (fixed a) moves toward the Rayleigh anomaly to spectral positions over which the lattice sum, and also its derivative, take larger values. From a physical point of view, this means that, as the lattice resonance gets closer to the Rayleigh anomaly, its collective character increases, and therefore, it becomes more sensitive to variations in wavelength, which, hence, results in a smaller linewidth.^{19,25}

All of the results discussed so far have focused on the far-field response of the array, that is, reflectance and absorbance, rather than on the dipole induced in the particles of the array. Examining this quantity, however, provides a useful insight into the nature of the lattice resonances induced by the vacancies. To that end, in Figure 4, we plot the dipoles induced in the particles of the unit cell of all of the arrays considered in Figure 3. Specifically, the color of the dots represents the value of the imaginary part of the x -component of the dipole induced at each particle at the first lattice resonance $\text{Im}\{p_{i,x}\}$, normalized to the largest absolute value in each unit cell. Notice that, for the first lattice resonance, this is the only significant component since we assume the array to be excited at normal incidence with an x -polarized electric field.⁵² Furthermore, although, the wavelengths at which the reflectance and absorbance are maximum do not always exactly coincide, the dipole plots at these two wavelengths are identical. The top row corresponds to arrays with constant interparticle distance $a_0 = 200$ nm, while the bottom row shows the behavior for constant period $a = 1400$ nm. In both cases, the nanoparticles have a diameter $D = 100$ nm and N varies from 2 (leftmost panel) to 7 (rightmost panel). The gray shaded areas indicate the unit cell.

To put this in context, under normal incidence excitation, all of the dipoles induced in the particles of a periodic array with a single-particle unit cell, such as the pristine or the

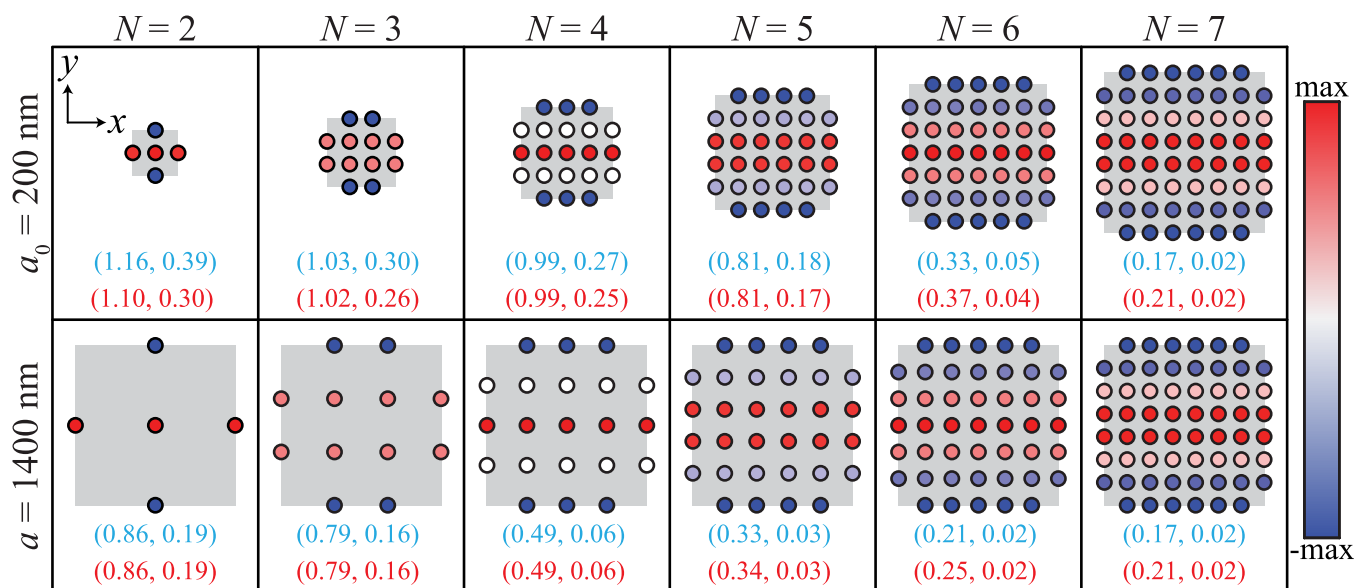


Figure 4. Analysis of the induced dipoles associated with the first lattice resonance of different arrays with vacancies. The color of the dots represents the value of $\text{Im}\{p_{ix}\}$ induced at each particle of the unit cell at the lattice resonance, normalized to the largest absolute value for the corresponding unit cell. The top and bottom rows correspond, respectively, to arrays with fixed interparticle distance $a_0 = 200 \text{ nm}$ and fixed period $a = 1400 \text{ nm}$. In both cases, N varies from 2 (leftmost panel) to 7 (rightmost panel), while the particle diameter is always $D = 100 \text{ nm}$. The unit cell is indicated by the gray shaded area and the numbers at the bottom of each panel correspond to $\text{Im}\{P\}/\Pi_0$ and $|P|^2/(2\Pi_0)^2$, where $P = \sum_{i=1}^{N^2-1} p_{ix}$ and $\Pi_0 = a^2 \lambda E_0 / (8\pi^2)$. These values are calculated at the wavelength for which the reflectance (blue text) and absorbance (red text) reach their maxima.

complementary arrays discussed in Figure 1, are identical.^{6,52,66} However, when vacancies are introduced to the array, this homogeneity is broken and the resulting dipoles display the patterns shown in Figure 4. These patterns can be understood from the fact that lattice resonances are the result of the far-field interaction between the particles of the array.⁵² For excitation at normal incidence with an x -polarized electric field, such a far-field interaction happens along the y -axis.⁵² This explains why the induced dipoles are uniform along any given row, but follow a nearly harmonic dependence with period a along the y -axis, reaching their largest magnitude at the rows containing the vacancies. This is especially visible in the $N = 4$ case, where the nodes and antinodes of the harmonic dependence coincide with the positions of the particles, resulting in two rows having a vanishing induced dipole and the others exhibiting maximum magnitude but with opposite sign. As expected, as the number of particles increases, the harmonic pattern across the unit cell becomes smoother because it is sampled at a larger number of points. However, the placement of additional particles in the unit cell becomes increasingly challenging due to the need to place them closer together.

The reflectance and absorbance of an array, for $\lambda > a$, under the illumination conditions considered here, can be written in terms of the total induced dipole of the unit cell, $P = \sum_{i=1}^{N^2-1} p_{ix}$ as²⁵

$$\mathcal{R} = |P|^2 / (2\Pi_0)^2, \quad \mathcal{A} = \text{Im}\{P\} / \Pi_0 - 2|P|^2 / (2\Pi_0)^2 \quad (3)$$

with $\Pi_0 = a^2 \lambda E_0 / (8\pi^2)$ (see the Supporting Information for a detailed derivation). These expressions provide a useful connection between the dipoles induced in the unit cell and the values of the reflectance and absorbance discussed in Figure 3. In order to exploit this connection, we print, at the bottom of each panel of Figure 4, the values of $\text{Im}\{P\}/\Pi_0$ and

$|P|^2/(2\Pi_0)^2$ calculated at the wavelengths that yield the maximum reflectance (blue text) and absorbance (red text) for each array. Examining these numbers, we observe that, as N increases, both $\text{Im}\{P\}/\Pi_0$ and $|P|^2/(2\Pi_0)^2$ decrease, which is consistent with the general trend observed in Figure 3. However, while the decrease of $|P|^2/(2\Pi_0)^2$ is, in all cases, very gradual, $\text{Im}\{P\}/\Pi_0$ drops off abruptly between $N = 5$ and $N = 6$ for the arrays with fixed interparticle distance a_0 (top row), thus explaining the sudden change in absorbance seen in Figure 3a.

Therefore, from the results shown in Figure 4, we conclude that the origin of the lattice resonances of these arrays can be traced back to the symmetry breaking introduced in the unit cell by the vacancies. Specifically, the presence of the vacancies results in a nonuniform dipole pattern in which the dipoles induced in the particles located at different rows oscillate with different phases, thus partially canceling each other, but still producing a nonvanishing total dipole. This subradiant character explains the larger absorbance and smaller reflectance that arrays with vacancies display as compared with their complementary counterparts.

Due to their simplicity, which facilitates the understanding of their optical response, we have focused on arrays with vacancies made of silver nanospheres that are surrounded by vacuum. However, any experimental verification of our predictions would require the array to be deposited on or embedded in a material. For that reason, in Figure 5, we propose and characterize a more experimentally feasible array with vacancies made of silver nanocylinders embedded in a uniform dielectric medium with refractive index $n = 1.46$, such as silica. In particular, the array has $N = 2$ and period $a = 700/n \text{ nm}$, while the nanocylinders have a height of 75 nm and a diameter of 100 nm, with corners rounded by a radius of curvature of 2 nm, as shown in the inset of panel (a) (see

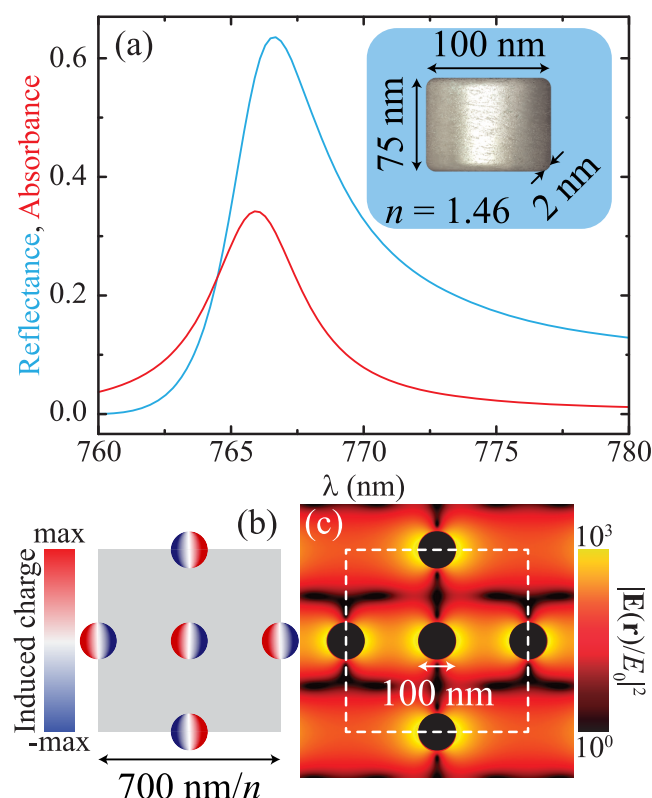


Figure 5. Potential implementation of an array with defects. (a) Reflectance (blue curve) and absorbance (red curve) of a $N = 2$ array of silver nanocylinders of height 75 nm and diameter 100 nm with corners rounded by a radius of curvature of 2 nm, as indicated by the inset. The cylinders are embedded in a medium with $n = 1.46$ and the period of the array is $a = 700/n$ nm. (b, c) Induced charge (b) on the surface of the nanocylinders and field intensity enhancement $|E(r)/E_0|^2$ (c) in a slice across the center of the nanocylinders. The gray shaded area in (b) and the white dashed box in (c) signal the unit cell. Both plots are calculated at a wavelength of 766 nm.

Figure S6 of the Supporting Information for an analysis of similar arrays with different periods and nanocylinder sizes). Under normal incidence illumination, we expect this array to display its first lattice resonance on the red side of $\lambda = 700$ nm. This prediction is confirmed in Figure 5a, where we plot the reflectance (blue curve) and absorbance (red curve) of the array, calculated using an FEM solver. Clearly, the array supports a strong lattice resonance induced by the periodic vacancies, which produces a peak reflectance of ~ 0.64 and a peak absorbance of ~ 0.34 . To gain further insight, we analyze in Figure 5b the charge distributions induced in each of the particles of the unit cell at the wavelength 766 nm. All of the nanodisks display the charge distribution of a horizontal dipole. However, the dipole of the particles in the top and bottom rows are in antiphase with respect to those of the particles in the middle row, in perfect agreement with the results for the spheres shown in Figure 4. We also analyze the enhancement of the field intensity $|E(r)/E_0|^2$ generated by the lattice resonance. Specifically, Figure 5c shows this quantity plotted using a logarithmic scale over a slice taken across the center of the nanocylinders, with the white dashed box outlining the unit cell. The enhancement of the field intensity displays a similar pattern around all of the particles, reaching a maximum value of $\sim 10^3$ at 5 nm from their surfaces. Interestingly, the enhancement shows significant values around the location of

the vacancies, reaching $\sim 10^2$ at their center, which indicates that any photonic elements placed at these positions will strongly interact with the lattice resonance of the array.

CONCLUSIONS

In summary, we have analyzed the optical response of periodic arrays of nanoparticles with periodic vacancies. Using a coupled dipole approach, we have shown that the introduction of these vacancies induces lattice resonances at wavelengths that are commensurate with their periodicity. Through a comparative analysis with the response of the complementary system consisting of the array formed by the particles removed from the system, we have found that the lattice resonances of the array with vacancies appear farther from the Rayleigh anomaly and produce a larger absorbance and a smaller reflectance than those of the complementary array. However, by adjusting the particle size and the array period, it is possible to alter the balance between reflectance and absorbance. Furthermore, we have studied the role played by the number of nanoparticles per unit cell in the optical response of these arrays, finding that the density of nanoparticles changes the strength, spectral position, and linewidth of the lattice resonance. By analyzing the dipoles induced in each of the particles of the unit cell at the lattice resonance, we have found that these modes arise from a symmetry breaking introduced by the presence of the vacancies, which results in a subradiant dipole pattern. Finally, we have proposed and characterized a potential implementation of an array with vacancies composed of silver nanocylinders embedded in a homogeneous dielectric environment. Our results provide insight into the nature and properties of the lattice resonances supported by arrays of metallic nanoparticles with complex unit cells. In particular, they can increase the potential of these systems for developing novel technological applications, such as nanolasers and ultrasensitive biosensors, that can benefit from the extra degrees of control provided by the presence of periodic vacancies.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsphotonics.0c01672>.

Derivation of eq 3; spectral dependence of nanosphere polarizability; comparison of the coupled dipole model with rigorous solutions of Maxwell's equations obtained using the finite element method; expanded reflectance and absorbance spectra for Figure 1; absorbance and reflectance spectra for varying numbers of nanoparticles per unit cell; analysis of potential implementations with different array periodicities and nanocylinder sizes (PDF)

AUTHOR INFORMATION

Corresponding Author

Alejandro Manjavacas – Department of Physics and Astronomy, University of New Mexico, Albuquerque, New Mexico 87106, United States; Instituto de Óptica - CSIC, 28006 Madrid, Spain; orcid.org/0000-0002-2379-1242; Email: manjavacas@unm.edu

Authors

Lauren Zundel – Department of Physics and Astronomy,
University of New Mexico, Albuquerque, New Mexico 87106,
United States; orcid.org/0000-0003-1850-5210

Asher May – Department of Physics and Astronomy,
University of New Mexico, Albuquerque, New Mexico 87106,
United States

Complete contact information is available at:

<https://pubs.acs.org/10.1021/acsp Photonics.0c01672>

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

This work has been partially supported by the U.S. National Science Foundation (Grant ECCS-1710697). We would like to thank the UNM Center for Advanced Research Computing, supported in part by the U.S. National Science Foundation, for providing part of the computational resources used in this work. L.Z. acknowledges support from the New Mexico Space Grant Consortium and the Department of Energy Computational Science Graduate Fellowship (Grant DE-SC0020347). We are also deeply grateful to Prof. Femius Koenderink for valuable and enjoyable discussions that inspired this work.

REFERENCES

- (1) Maier, S. A. *Plasmonics: Fundamentals and Applications*; Springer: New York, 2007.
- (2) Álvarez-Puebla, R. A.; Liz-Marzán, L. M.; García de Abajo, F. J. Light concentration at the nanometer scale. *J. Phys. Chem. Lett.* **2010**, *1*, 2428–2434.
- (3) Halas, N. J.; Lal, S.; Chang, W.; Link, S.; Nordlander, P. Plasmons in strongly coupled metallic nanostructures. *Chem. Rev.* **2011**, *111*, 3913–3961.
- (4) Zhao, L.; Kelly, K. L.; Schatz, G. C. The extinction spectra of silver nanoparticle arrays: influence of array structure on plasmon resonance wavelength and width. *J. Phys. Chem. B* **2003**, *107*, 7343–7350.
- (5) Zou, S.; Janel, N.; Schatz, G. C. Silver nanoparticle array structures that produce remarkably narrow plasmon lineshapes. *J. Chem. Phys.* **2004**, *120*, 10871–10875.
- (6) García de Abajo, F. J. Colloquium: Light scattering by particle and hole arrays. *Rev. Mod. Phys.* **2007**, *79*, 1267–1290.
- (7) Auguie, B.; Barnes, W. L. Collective resonances in gold nanoparticle arrays. *Phys. Rev. Lett.* **2008**, *101*, 143902.
- (8) Kravets, V. G.; Schedin, F.; Grigorenko, A. N. Extremely narrow plasmon resonances based on diffraction coupling of localized plasmons in arrays of metallic nanoparticles. *Phys. Rev. Lett.* **2008**, *101*, 087403.
- (9) Chu, Y.; Schonbrun, E.; Yang, T.; Crozier, K. B. Experimental observation of narrow surface plasmon resonances in gold nanoparticle arrays. *Appl. Phys. Lett.* **2008**, *93*, 181108.
- (10) Cherqui, C.; Bourgeois, M. R.; Wang, D.; Schatz, G. C. Plasmonic Surface Lattice Resonances: Theory and Computation. *Acc. Chem. Res.* **2019**, *52*, 2548–2558.
- (11) Wang, W.; Ramezani, M.; Väkeväinen, A. I.; Törmä, P.; Gómez Rivas, J.; Odom, T. W. The rich photonic world of plasmonic nanoparticle arrays. *Mater. Today* **2018**, *21*, 303–314.
- (12) Kravets, V. G.; Kabashin, A. V.; Barnes, W. L.; Grigorenko, A. N. Plasmonic Surface Lattice Resonances: A Review of Properties and Applications. *Chem. Rev.* **2018**, *118*, 5912–5951.
- (13) Vecchi, G.; Giannini, V.; Gómez Rivas, J. Surface modes in plasmonic crystals induced by diffractive coupling of nanoantennas. *Phys. Rev. B: Condens. Matter Mater. Phys.* **2009**, *80*, 201401.
- (14) Teperik, T. V.; Degiron, A. Design strategies to tailor the narrow plasmon-photonic resonances in arrays of metallic nanoparticles. *Phys. Rev. B: Condens. Matter Mater. Phys.* **2012**, *86*, 245425.
- (15) Humphrey, A. D.; Barnes, W. L. Plasmonic surface lattice resonances on arrays of different lattice symmetry. *Phys. Rev. B: Condens. Matter Mater. Phys.* **2014**, *90*, 075404.
- (16) Nikitin, A. G.; Kabashin, A. V.; Dallaporta, H. Plasmonic resonances in diffractive arrays of gold nanoantennas: near and far field effects. *Opt. Express* **2012**, *20*, 27941–27952.
- (17) Vitrey, A.; Aigouy, L.; Prieto, P.; García-Martín, J. M.; González, M. U. Parallel Collective Resonances in Arrays of Gold Nanorods. *Nano Lett.* **2014**, *14*, 2079–2085.
- (18) Huang, Y.; Ma, L.; Hou, M.; Zhang, Z. Universal Near-Field Interference Patterns of Fano Resonances in Two-Dimensional Plasmonic Crystals. *Plasmonics* **2016**, *11*, 1377–1383.
- (19) Manjavacas, A.; Zundel, L.; Sanders, S. Analysis of the Limits of the Near-Field Produced by Nanoparticle Arrays. *ACS Nano* **2019**, *13*, 10682–10693.
- (20) Zhou, W.; Odom, T. W. Tunable subradiant lattice plasmons by out-of-plane dipolar interactions. *Nat. Nanotechnol.* **2011**, *6*, 423.
- (21) Zhao, W.; Jiang, Y. Experimental demonstration of sharp Fano resonance within binary gold nanodisk array through lattice coupling effects. *Opt. Lett.* **2015**, *40*, 93–96.
- (22) Le-Van, Q.; Zoethout, E.; Geluk, E.-J.; Ramezani, M.; Berghuis, M.; Gómez Rivas, J. Enhanced Quality Factors of Surface Lattice Resonances in Plasmonic Arrays of Nanoparticles. *Adv. Opt. Mater.* **2019**, *7*, 1801451.
- (23) Liu, S.-D.; Yue, P.; Zhang, S.; Wang, M.; Dai, H.; Chen, Y.; Nie, Z.-Q.; Cui, Y.; Han, J.-B.; Duan, H. Metasurfaces Composed of Plasmonic Molecules: Hybridization Between Parallel and Orthogonal Surface Lattice Resonances. *Adv. Opt. Mater.* **2020**, *8*, 1901109.
- (24) Bin-Alam, M. S.; Reshef, O.; Mamchur, Y.; Alam, M. Z.; Carlow, G.; Upham, J.; Sullivan, B. T.; Ménard, J.-M.; Huttunen, M. J.; Dolgaleva, K.; Boyd, R. W. Ultra-High-Q Resonances in Plasmonic Metasurfaces. *arXiv* **2020**, 05202.
- (25) Cuartero-González, A.; Sanders, S.; Zundel, L.; Fernández-Domínguez, A. I.; Manjavacas, A. Super- and Subradiant Lattice Resonances in Bipartite Nanoparticle Arrays. *ACS Nano* **2020**, *14*, 11876.
- (26) Adato, R.; Yanik, A. A.; Amsden, J. J.; Kaplan, D. L.; Omenetto, F. G.; Hong, M. K.; Erramilli, S.; Altug, H. Ultra-sensitive vibrational spectroscopy of protein monolayers with plasmonic nanoantenna arrays. *Proc. Natl. Acad. Sci. U. S. A.* **2009**, *106*, 19227–19232.
- (27) Kravets, V. G.; Schedin, F.; Jalil, R.; Britnell, L.; Gorbachev, R. V.; Ansell, D.; Thackray, B.; Novoselov, K. S.; Geim, A. K.; Kabashin, A. V.; Grigorenko, A. N. Singular phase nano-optics in plasmonic metamaterials for label-free single-molecule detection. *Nat. Mater.* **2013**, *12*, 304–309.
- (28) Thackray, B. D.; Kravets, V. G.; Schedin, F.; Auton, G.; Thomas, P. A.; Grigorenko, A. N. Narrow Collective Plasmon Resonances in Nanostructure Arrays Observed at Normal Light Incidence for Simplified Sensing in Asymmetric Air and Water Environments. *ACS Photonics* **2014**, *1*, 1116–1126.
- (29) Danilov, A.; Tselikov, G.; Wu, F.; Kravets, V. G.; Ozerov, I.; Bedu, F.; Grigorenko, A. N.; Kabashin, A. V. Ultra-narrow surface lattice resonances in plasmonic metamaterial arrays for biosensing applications. *Biosens. Bioelectron.* **2018**, *104*, 102–112.
- (30) Rodriguez, S. R. K.; Lozano, G.; Verschuuren, M. A.; Gomes, R.; Lambert, K.; De Geyter, B.; Hassinen, A.; Van Thourhout, D.; Hens, Z.; Gomez Rivas, J. Quantum rod emission coupled to plasmonic lattice resonances: A collective directional source of polarized light. *Appl. Phys. Lett.* **2012**, *100*, 111103.
- (31) Lozano, G.; Louwers, D. J.; Rodríguez, S. R. K.; Murai, S.; Jansen, O. T. A.; Verschuuren, M. A.; Gómez, J. Plasmonics for solid-state lighting: enhanced excitation and directional emission of highly efficient light sources. *Light: Sci. Appl.* **2013**, *2*, e66.
- (32) Zhou, W.; Dridi, M.; Suh, J. Y.; Kim, C. H.; Co, D. T.; Wasielewski, M. R.; Schatz, G. C.; Odom, T. W. Lasing action in

strongly coupled plasmonic nanocavity arrays. *Nat. Nanotechnol.* **2013**, *8*, 506–511.

(33) Schokker, A. H.; Koenderink, A. F. Lasing at the band edges of plasmonic lattices. *Phys. Rev. B: Condens. Matter Mater. Phys.* **2014**, *90*, 155452.

(34) Cotrufo, M.; Osorio, C. I.; Koenderink, A. F. Spin-Dependent Emission from Arrays of Planar Chiral Nanoantennas Due to Lattice and Localized Plasmon Resonances. *ACS Nano* **2016**, *10*, 3389–3397.

(35) Hakala, T. K.; Rekola, H. T.; Väkeväinen, A. I.; Martikainen, J.-P.; Nečada, M.; Moilanen, A. J.; Törmä, P. Lasing in dark and bright modes of a finite-sized plasmonic lattice. *Nat. Commun.* **2017**, *8*, 13687.

(36) Schokker, A. H.; van Riggelen, F.; Hadad, Y.; Alù, A.; Koenderink, A. F. Systematic study of the hybrid plasmonic-photonic band structure underlying lasing action of diffractive plasmon particle lattices. *Phys. Rev. B: Condens. Matter Mater. Phys.* **2017**, *95*, 085409.

(37) Wang, D.; Hua, W.; Wang, A.; Hua, R. D.; Schaller, Y.; Schatz, G. C.; Odom, T. W. Band-edge engineering for controlled multi-modal nanolasing in plasmonic superlattices. *Nat. Nanotechnol.* **2017**, *12*, 889.

(38) Guo, R.; Nečada, M.; Hakala, T. K.; Väkeväinen, A. I.; Törmä, P. Lasing at K Points of a Honeycomb Plasmonic Lattice. *Phys. Rev. Lett.* **2019**, *122*, 013901.

(39) Hu, J.; Wang, D.; Bhowmik, D.; Liu, T.; Deng, S.; Knudson, M. P.; Ao, X.; Odom, T. W. Lattice-Resonance Metalenses for Fully Reconfigurable Imaging. *ACS Nano* **2019**, *13*, 4613–4620.

(40) Czaplicki, R.; Kiviniemi, A.; Huttunen, M. J.; Zang, X.; Stolt, T.; Vartiainen, I.; Butet, J.; Kuittinen, M.; Martin, O. J. F.; Kauranen, M. Less Is More: Enhancement of Second-Harmonic Generation from Metasurfaces by Reduced Nanoparticle Density. *Nano Lett.* **2018**, *18*, 7709–7714.

(41) Olson, J.; Manjavacas, A.; Basu, T.; Huang, D.; Schlather, A. E.; Zheng, B.; Halas, N. J.; Nordlander, P.; Link, S. High chromaticity aluminum plasmonic pixels for active liquid crystal displays. *ACS Nano* **2016**, *10*, 1108–1117.

(42) Kristensen, A.; Yang, J. K. W.; Bozhevolnyi, S. I.; Link, S.; Nordlander, P.; Halas, N. J.; Mortensen, N. A. Plasmonic colour generation. *Nat. Rev. Mater.* **2017**, *2*, 16088.

(43) Esposito, M.; Todisco, F.; Bakhti, S.; Passaseo, A.; Tarantini, I.; Cuscunà, M.; Destouches, N.; Tasco, V. Symmetry Breaking in Oligomer Surface Plasmon Lattice Resonances. *Nano Lett.* **2019**, *19*, 1922–1930.

(44) Rodríguez, S. R. K.; Feist, J.; Verschuuren, M. A.; Garcia Vidal, F. J.; Gómez Rivas, J. Thermalization and Cooling of Plasmon-Exciton Polaritons: Towards Quantum Condensation. *Phys. Rev. Lett.* **2013**, *111*, 166802.

(45) Väkeväinen, A. I.; Moerland, R. J.; Rekola, H. T.; Eskelinen, A.-P.; Martikainen, J.-P.; Kim, D.-H.; Törmä, P. Plasmonic Surface Lattice Resonances at the Strong Coupling Regime. *Nano Lett.* **2014**, *14*, 1721–1727.

(46) Hakala, T. K.; Moilanen, A. J.; Väkeväinen, A. I.; Guo, R.; Martikainen, J.-P.; Daskalakis, K. S.; Rekola, H. T.; Julku, A.; Törmä, P. Bose–Einstein condensation in a plasmonic lattice. *Nat. Phys.* **2018**, *14*, 739–744.

(47) Kwadrin, A.; Koenderink, A. F. Diffractive stacks of metamaterial lattices with a complex unit cell: Self-consistent long-range bianisotropic interactions in experiment and theory. *Phys. Rev. B: Condens. Matter Mater. Phys.* **2014**, *89*, 045120.

(48) Lunnemann, P.; Koenderink, A. F. Dispersion of guided modes in two-dimensional split ring lattices. *Phys. Rev. B: Condens. Matter Mater. Phys.* **2014**, *90*, 245416.

(49) Wang, D.; Yang, A.; Hryn, A. J.; Schatz, G. C.; Odom, T. W. Superlattice Plasmons in Hierarchical Au Nanoparticle Arrays. *ACS Photonics* **2015**, *2*, 1789–1794.

(50) Humphrey, A. D.; Barnes, W. L. Plasmonic surface lattice resonances in arrays of metallic nanoparticle dimers. *J. Opt.* **2016**, *18*, 035005.

(51) Guo, R.; Hakala, T. K.; Törmä, P. Geometry dependence of surface lattice resonances in plasmonic nanoparticle arrays. *Phys. Rev. B: Condens. Matter Mater. Phys.* **2017**, *95*, 155423.

(52) Baur, S.; Sanders, S.; Manjavacas, A. Hybridization of Lattice Resonances. *ACS Nano* **2018**, *12*, 1618–1629.

(53) Kolkowski, R.; Koenderink, A. F. Lattice Resonances in Optical Metasurfaces With Gain and Loss. *Proc. IEEE* **2019**, 1–24.

(54) Li, R.; Bourgeois, M. R.; Cherqui, C.; Guan, J.; Wang, D.; Hu, J.; Schaller, R. D.; Schatz, G. C.; Odom, T. W. Hierarchical Hybridization in Plasmonic Honeycomb Lattices. *Nano Lett.* **2019**, *19*, 6435–6441.

(55) Fradkin, I. M.; Dyakov, S. A.; Gippius, N. A. Nanoparticle lattices with bases: Fourier modal method and dipole approximation. *Phys. Rev. B: Condens. Matter Mater. Phys.* **2020**, *102*, 045432.

(56) Becerril, D.; Vázquez, O.; Piccotti, D.; Sandoval, E. M.; Cesca, T.; Mattei, G.; Noguez, C.; Pirruccio, G. Diffractive dipolar coupling in non-Bravais plasmonic lattices. *Nanoscale Adv.* **2020**, *2*, 1261–1268.

(57) Abujetas, D. R.; Olmos-Trigo, J.; Sáenz, J. J.; Sánchez-Gil, J. A. Coupled electric and magnetic dipole formulation for planar arrays of particles: Resonances and bound states in the continuum for all-dielectric metasurfaces. *Phys. Rev. B: Condens. Matter Mater. Phys.* **2020**, *102*, 125411.

(58) Humphrey, A. D.; Meinzer, N.; Starkey, T. A.; Barnes, W. L. Surface Lattice Resonances in Plasmonic Arrays of Asymmetric Disc Dimers. *ACS Photonics* **2016**, *3*, 634–639.

(59) Schaafsma, M. C.; Bhattacharya, A.; Gómez Rivas, J. Diffraction Enhanced Transparency and Slow THz Light in Periodic Arrays of Detuned and Displaced Dipoles. *ACS Photonics* **2016**, *3*, 1596–1603.

(60) Abujetas, D. R.; van Hoof, N.; ter Huurne, S.; Rivas, J. G.; Sánchez-Gil, J. A. Spectral and temporal evidence of robust photonic bound states in the continuum on terahertz metasurfaces. *Optica* **2019**, *6*, 996–1001.

(61) Moon, K.; Lee, S.; Han, H. Nearly flat donor and acceptor bands of two-dimensional photonic crystals with periodic vacancy defects. *J. Korean Phys. Soc.* **2007**, *50*, 532–536.

(62) Hack, S. A.; van der Vegt, J. J. W.; Vos, W. L. Cartesian light: Unconventional propagation of light in a three-dimensional superlattice of coupled cavities within a three-dimensional photonic band gap. *Phys. Rev. B: Condens. Matter Mater. Phys.* **2019**, *99*, 115308.

(63) García de Abajo, F. J. Multiple scattering of radiation in clusters of dielectrics. *Phys. Rev. B: Condens. Matter Mater. Phys.* **1999**, *60*, 6086–6102.

(64) Johnson, P. B.; Christy, R. W. Optical constants of the noble metals. *Phys. Rev. B* **1972**, *6*, 4370–4379.

(65) Thongrattanasiri, S.; Koppens, F. H. L.; García de Abajo, F. J. Complete optical absorption in periodically patterned graphene. *Phys. Rev. Lett.* **2012**, *108*, 047401.

(66) Zundel, L.; Manjavacas, A. Finite-size effects on periodic arrays of nanostructures. *J. Phys.: Photonics* **2019**, *1*, 015004.